Algebraic substitution

Expression substitution - mixed practice

Mix substitution into a single expression with collecting like terms.

10 MINUTES / 4 QUESTIONS / AS-11

Name _____ Date _____

Show your working. Use brackets for negative values.

Starter: use the full set. Exit ticket: choose 2-3 questions.

1 Given $x = 5y - 5$

Write in terms of y and simplify

Simplify $5x + 5$

2 Given $a = 6b - 6$

Write in terms of b and simplify

Simplify $4a + 2b$

3 Given $x = 3y + 5$

Write in terms of y and simplify

Simplify $3x$

4 Given $a = 5b + 6$

Write in terms of b and simplify

Simplify $5a + 4b$

Answer sheet

Expression substitution - mixed practice | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $25y - 20$

02 $26b - 24$

03 $9y + 15$

04 $29b + 30$

Worked solutions

Expression substitution - mixed practice

01 Given $x = 5y - 5$

$5x + 5$

Substitute $x = 5y - 5$.

$5(5y - 5) + 5$

$25y - 25 + 5$

$25y - 20$

02 Given $a = 6b - 6$

$4a + 2b$

Substitute $a = 6b - 6$.

$4(6b - 6) + 2b$

$24b - 24 + 2b$

$(24 + 2)b - 24 = 26b - 24$

03 Given $x = 3y + 5$

$3x$

Substitute $x = 3y + 5$.

$3(3y + 5)$

$9y + 15$

$9y + 15$

04 Given $a = 5b + 6$

$5a + 4b$

Substitute $a = 5b + 6$.

$5(5b + 6) + 4b$

$25b + 30 + 4b$

$(25 + 4)b + 30 = 29b + 30$